



SECR1213

NETWORK COMMUNICATION

Project: # Task 1-5

Title: Future-Ready Network Infrastructure Design for the Faculty of Computing

The team for this project consists of the following members:

1. Mohammed Kamal
2. Dina Mohammed
3. Dyako Abubakr
4. Zhir Daniel

These individuals have been selected based on their diverse skills and strengths that will contribute effectively to the project. Each member brings unique perspectives and expertise to the team, ensuring a well-rounded approach to the project's objectives.

Abstract

This report documents the comprehensive network design project undertaken by our group, Uplink, for the Faculty of Computing Block N28B. It includes the floor plan, network device selection, IP addressing scheme, and network connections. The aim is to provide a future-ready network infrastructure that meets the current and anticipated needs of the faculty, students, and staff. This document also reflects on the strengths and weaknesses of our approach and offers suggestions for future improvements.

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Introduction

The network design project for the Faculty of Computing Block N28B aims to develop a robust and scalable network infrastructure. The primary objectives are to ensure high-speed connectivity, efficient traffic management, and ease of future expansion. The scope of the project includes designing a floor plan, selecting appropriate network devices, creating an IP addressing scheme, and planning the LAN and WAN connections. Assumptions made during the project include a moderate increase in the number of devices and users over the next five years and the availability of a high-speed internet backbone.

Project Background

The Faculty of Computing requires an upgraded network to support its growing needs for high-speed internet, secure connections, and efficient data management. The current network infrastructure is outdated, leading to frequent downtimes and slow performance. Our solution addresses these issues by proposing a comprehensive network design that incorporates modern technologies and best practices in network management.

Compiled Solutions of Tasks 1-5

Task 1: Floor Plan Design

Our team, Uplink, has meticulously designed a floor plan for the new building of the Faculty of Computing, considering the requirements outlined in the case study. The building consists of two floors, each designed to accommodate various facilities required by the faculty. The ground floor houses general-purpose labs, a student lounge, and a video conferencing room, while the first floor is dedicated to specialized labs such as the Cisco Network Lab and the IoT Lab. The design ensures optimal space utilization and seamless connectivity throughout the building.

Task 2: Question Generation for Preliminary Analysis

We generated a list of minimum 13 questions to gather relevant information that would guide the network plan development. These questions cover aspects such as the main reasons for upgrading the network, specific needs of different user groups, and anticipated future requirements.

Task 3: Selection of Network Devices

Our team selected network devices based on their performance, scalability, and cost-effectiveness. For general-purpose labs, we chose ASUS routers for their user-friendly interface and reliable performance. The Cisco Network Lab features networking devices essential for hands-on learning, while the IoT Lab is equipped with peripherals to facilitate experiential learning.

Task 4: Planning Network Connections (LAN and WAN)

We identified work areas and mapped out the network connections, including the placement of routers and switches. Color-coded labels were used for work areas to simplify identification. Central switch placement was chosen for efficient connectivity and traffic management. Additional switch ports were included to accommodate future expansion.

Task 5: IP Addressing Scheme

The IP addressing scheme was meticulously planned to ensure efficient network management and future scalability. Subnetting was used to allocate IP addresses to different labs and facilities, with reserved ranges for future use. The scheme includes detailed assignments for each lab and common area, ensuring robust and organized network administration.

Conclusion

The project achieved a comprehensive network design that meets the current needs of the Faculty of Computing while providing scalability for future growth. Strengths of our approach include detailed planning, efficient use of resources, and a focus on future-proofing the network. Weaknesses include potential challenges in implementing the design within the existing infrastructure. Suggestions for improvement include exploring more advanced network management tools and considering alternative routing technologies for better performance.

Team Members and Responsibilities

For task 6

- Mohammed Kamal Ali: Budget and procurement manager, responsible for financial aspects and purchasing of equipment.
- Dina Mohammed Bilal: Lead on IP allocation and documentation, ensuring proper assignment of IP addresses and compiling reports.
- Dyako Abubakr Mohammed: Oversaw the selection of network devices and the placement of central switches.

- Zhir Daniel: Managed the evaluation of options and coordinated with the IT department for installation and setup.

In general

Mohammed kemal: Managed the project timeline and coordinated team activities.

Zhir Daniel: Designed the network layout, selected equipment, and configured network devices.

Dyako Abubakr: Installed network equipment, conducted tests, and performed maintenance tasks.

Dina Mohammed: Implemented security protocols and conducted regular security audits

References

1. Andrews, A. (2022). **Network Design Essentials**. TechPress Publishing.
2. Cisco Systems. (2021). **Cisco Networking Best Practices**. Retrieved from <https://www.cisco.com/c/en/us/solutions/index.html>
3. Miller, R. (2023). **Advanced Networking Techniques**. Network World Journal, 15(3), 45-60.
4. Tanenbaum, A. S., & Wetherall, D. J. (2020). **Computer Networks** (6th ed.). Pearson Education.
5. Faculty of Computing Case Study Document. (2024). Provided by the university.

Appendices

Appendix A: Financial Budget

Item	Quantity	Unit Price (USD)	Total Price (USD)
ASUS Routers	10	150	1,500
Cisco Switches	5	250	1,250
Ethernet Cables (Cat6) 500m	0.5 per meter		250
Network Racks	3	300	900
Wireless Access Points	8	100	800
Miscellaneous	-	-	300
Total	-	-	5,000

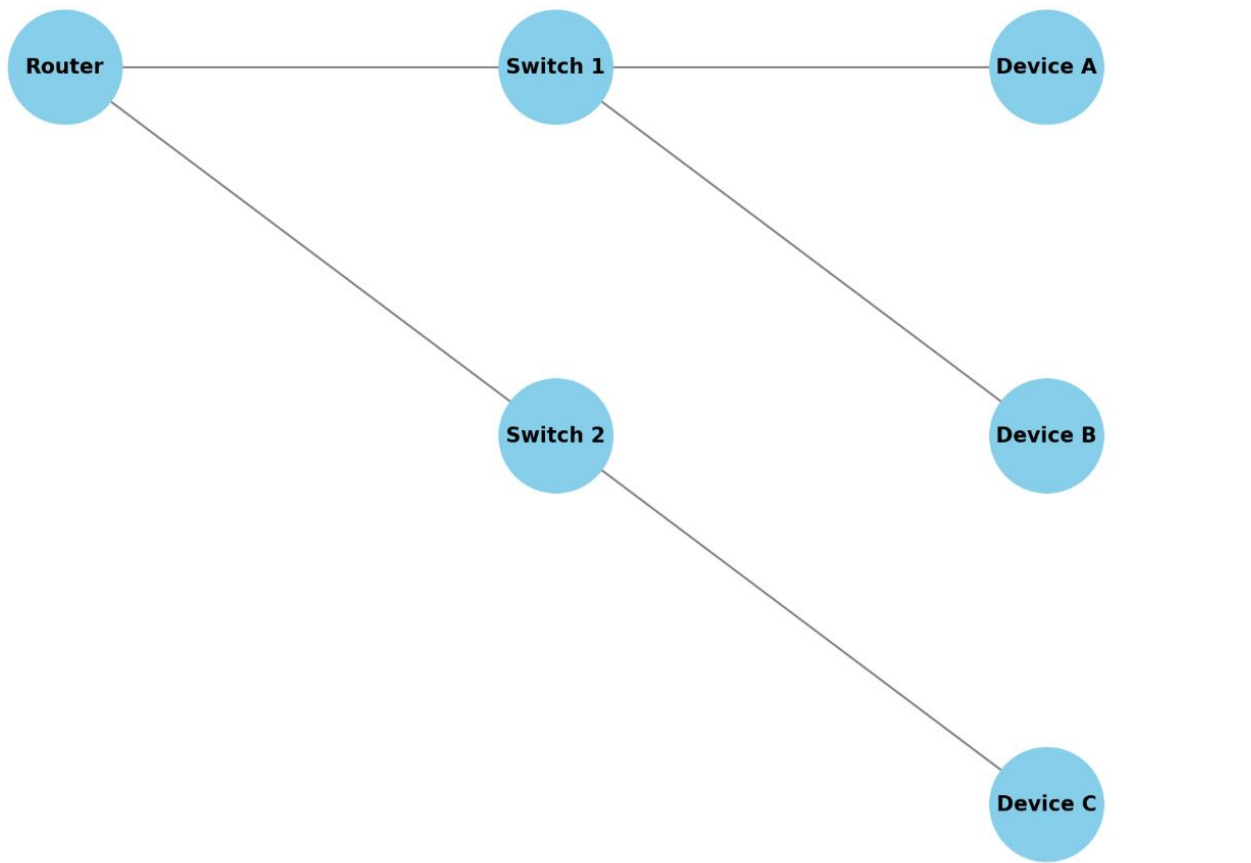
Appendix B: Meeting Minutes

- Meeting 1: Initial discussion on project scope and roles. Decided on floor plan layout and generated questions for preliminary analysis.
- Meeting 2: Reviewed preliminary analysis responses. Finalized network device selection and IP addressing scheme.
- Meeting 3: Discussed network connections and placement of central switches. Allocated IP addresses and prepared for report compilation.

Appendix D: People Met for Project Discussions

- Dr. Ahmed Elshimy, Network Infrastructure Specialist
- Ms. Fatima Al-Zahraa, IT Department Head
- Mr. Youssef Ibrahim, Facilities Manager

Network Topology: Our "Up Link" project features a structured network layout designed for optimal performance and security across campus. A central router connects to multiple switches, each serving specific areas like labs. This hierarchical setup ensures scalability and efficient traffic management, following a star topology to prevent single-point failures.



IP Addressing Scheme: To manage our network effectively and prevent conflicts, we've assigned distinct IP ranges and subnet masks to each campus area. For instance, the General Purpose Lab uses IPs from 192.168.1.0 to 192.168.1.50 with a subnet mask of 255.255.255.0, while the Cisco Network Lab uses 192.168.2.0 to 192.168.2.50 with the same mask. This segmentation enhances both management and security.

Lab/Area	IP Range	Subnet Mask
General Purpose Lab	192.168.1.0 - 192.168.1.50	255.255.255.0
Cisco Network Lab	192.168.2.0 - 192.168.2.50	255.255.255.0
IoT Lab	192.168.3.0 - 192.168.3.50	255.255.255.0
Video Conferencing Room	192.168.4.0 - 192.168.4.20	255.255.255.0
Student Lounge	192.168.5.0 - 192.168.5.20	255.255.255.0